Practical 9:

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
~
  GNU nano 8.7
int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], n=12, frames=3;
    int i,j,k,hit=0,fault=0,pos,far;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    frame[j] = pages[i];
                    found = 1;
                    break;
                }
            }

            if(!found) {
                far = -1;
                pos = -1;

                for(j=0;j<frames;j++) {
                    int next = -1;

                    for(k=i+1;k<n;k++) {
                        if(frame[j] == pages[k]) {
                            next = k;
                            break;
                        }
                    }

                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > far) {
                        far = next;
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}
```

Output:

```
LENOVO@shravil MSYS ~
$ ./practical_9.exe
Enter number of pages: 1 2 3 4 1 2 5 1 2 3 4 5
Enter page reference string:
Enter number of frames: Hits = 0
Faults = 1
Hit % = 0.00
Fault % = 100.00
```